

AB \ CD	00	01	11	10
00	1			X
01		X		1
11				1
10	1			X

$$F = B'D' + CD'$$

3-16: Simplify the boolean function F using the don't-care conditions d , in (1) sum of products and (2) product of sums:

$$(a) F = A'B'D' + A'CD + A'BC$$

$$d = A'BC'D + ACD + AB'D'$$

$$F = A'B'C'D' + A'BCD' + A'B'CD + A'BCD + A'BCD'$$

$$F = \Sigma(0, 2, 3, 7, 6)$$

$$d = A'BC'D + A'B'CD + ABCD + AB'C'D' + AB'CD'$$

$$\Sigma(5, 8, 10, 11, 15)$$

$$F = A' + B'D'$$

AB \ CD	00	01	11	10
00	1		1	1
01		X	1	1
11			X	
10	X		X	X

$$F' = BC' + A + CD$$

$$F = (B' + C) \cdot A' \cdot (C + D')$$

AB \ CD	00	01	11	10
00		1		
01	1	X		
11	1	1	X	1
10	X	1	X	X

$$(c) F = ACE + A'CD'E' + AC'DE$$

$$d = DE + A'D'E' + AD'E'$$

$$F = AC'DE + A'CD'E' + A'CD'E + AC'DE$$

$$d = A'CD'E' + A'CD'E + AC'DE + AC'DE + A'CD'E + A'CD'E + A'CD'E + A'CD'E'$$

$$F = \Sigma(3, 4, 13, 15) \quad d = \Sigma(1, 2, 4, 5, 6, 8, 10, 12, 14)$$